

Ans 5. A force can change an object's speed, and it can change its direction of motion.

Ans 2. Newton's second law: the net force on a body equals mass times acceleration, $F = ma$.

Ans 1. The SI unit of force is the Newton (N).

Ans 4. $KE = \frac{1}{2} m v^2$.

Ans 3. Acceleration is the rate of change of velocity. SI unit: m/s^2 .